Quantum Gravitational Dynamics

Chapter 4: Equations of Motion

From the σ-Field to the Worldline

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March 2026

Abstract

We derive the equations of motion in Quantum Gravitational Dynamics at three levels of description: (i) the σ-field equation governing the entire system; (ii) the modified worldline equation for a test particle, which replaces the geodesic equation of General Relativity; and (iii) the conserved stress-energy equation encoding the interplay between matter, the σ-field, and the Pais–Uhlenbeck massive mode. The central structural result is the Advection Tensor $\Omega^{\lambda}_{\mu\nu}$, which arises from the composite nature of the QGD metric $g_{\mu\nu}(\sigma_t)$ and encodes the field-dragging effect absent in GR. We derive $\Omega^{\lambda}_{\mu\nu}$ exactly from the particle action, compute its explicit components in the Schwarzschild sector, and establish the precise sense in which it generalises and reduces to the standard Levi-Civita connection. The relationship between the three levels is made explicit, and a comparison table between GR and QGD motions is provided.

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1 Overview: Three Levels of Motion in QGD

In General Relativity the motion of matter is governed by two logically separate laws: the geodesic equation (worldline) and the Einstein field equations (geometry). In QGD these are unified. The metric is a composite functional of the σ-field, $g_{\mu\nu} = g_{\mu\nu}(\sigma_t)$, so the field equation and the worldline equation are both consequences of the same master action.

The equations of motion appear at three levels, each a projection of the same underlying field theory:

1. **Field level.** The Pais–Uhlenbeck master equation governs the σ-field sourced by all matter:
$$\Box_g\big(1 - \kappa\ell_Q^2\,\Box_g\big)\,\sigma_\mu = S_\mu, \qquad S_\mu = Q_\mu + G_\mu + T_\mu. \tag{1}$$
This is the most fundamental statement of motion in QGD.

2. **Worldline level.** For a test particle of mass μ, varying the particle action $S_{\rm pp} = -\mu\int ds$ with respect to the worldline $x^\mu(\lambda)$ in the composite metric $g_{\mu\nu}(\sigma_t)$ gives:
$$\frac{d^2x^\mu}{ds^2} + \Big(\Gamma^\mu_{\alpha\beta} + \Omega^\mu_{\alpha\beta}\Big)\frac{dx^\alpha}{ds}\frac{dx^\beta}{ds} = 0, \tag{2}$$
where $\Gamma^\mu_{\alpha\beta}$ is the standard Levi-Civita connection of the background field $g_{\mu\nu}(\sigma_2)$ and $\Omega^\mu_{\alpha\beta}$ is the *Advection Tensor* arising from the composite metric structure.

3. **Conservation level.** The stress-energy of the full system—matter, σ-field, and Pais–Uhlenbeck massive mode—is covariantly conserved:
$$\nabla_\nu\big(T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_{\rm field} + \Theta^{\mu\nu}_{\rm PU}\big) = 0. \tag{3}$$

In the limit $\ell_Q \to 0$ and for a static, weak σ-field, all three levels reduce to the standard GR results. The QGD corrections—the Advection Tensor at the worldline level and the PU stress tensor at the conservation level—become non-negligible only at post-Newtonian order or near the Planck scale.

2 The Field-Level Equation of Motion

2.1 Particles as field configurations

In QGD a particle of mass m_a is not a structureless point on a background geometry; it is identified with its own single-body σ-field configuration,

$$\sigma_t^{(a)}(\mathbf{x}) = \sqrt{\frac{r_s^{(a)}}{|\mathbf{x}-\mathbf{x}_a|}}, \qquad r_s^{(a)} \equiv \frac{2Gm_a}{c^2}. \tag{4}$$

This is the exact Schwarzschild solution for body a in isolation. The "position" of the particle is the center of mass of this field configuration, and its "motion" is the evolution of that center of mass as the field adjusts to the sources present.

The total field for an N-body system is the superposition plus interactions:

$$\sigma_t^{\text{tot}}(\mathbf{x}, t) = \sum_{a=1}^{N} \sigma_t^{(a)}(\mathbf{x} - \mathbf{x}_a(t)) + \delta\sigma(\mathbf{x}, \{\mathbf{x}_a\}, \{\mathbf{v}_a\}). \tag{5}$$

The interaction field $\delta\sigma$ is determined by the integral equation from Chapter 3: $\sigma = \sigma^{\text{free}} + \int [G_0 - G_{m_Q}] S$.

2.2 The self-consistent field equation

The motion of body a is encoded in the requirement that the master field equation (1) is satisfied at all times. Projected onto the worldline of body a, this becomes a self-consistency condition: the field $\sigma_t^{(a)}$ centred on $\mathbf{x}_a(t)$ must satisfy the field equation in the background of all other bodies. There is no separate "geodesic equation" at the fundamental level.

The trajectory $\mathbf{x}_a(t)$ is determined implicitly by

$$\Box_g \big(1 - \kappa \ell_Q^2 \, \Box_g\big) \, \sigma_t^{(a)} \big(\mathbf{x} - \mathbf{x}_a(t)\big) = -Q_\mu^{\text{cross}} \left[\sigma_t^{(a)}, \sum_{b \neq a} \sigma_t^{(b)} \right] + \cdots , \tag{6}$$

where Q_μ^{cross} is the cross-source from the nonlinear self-interaction (Section 4.1 of Chapter 3).

3 The Modified Geodesic Equation

3.1 Derivation from the particle action

For a test particle (a body whose own σ-field does not appreciably back-react on the background), the effective worldline is derived by varying the particle action

$$S_{\text{pp}} = -\mu \int \sqrt{-g_{\mu\nu}(\sigma_t) \, \dot{x}^\mu \, \dot{x}^\nu} \; d\lambda \tag{7}$$

with respect to $x^\mu(\lambda)$. The Euler–Lagrange equations give the standard-looking form with modified Christoffel symbols.

The crucial step is the evaluation of $\partial g_{\alpha\beta}/\partial x^\mu$. Since $g_{\alpha\beta} = g_{\alpha\beta}(\sigma_t)$ and $\sigma_t = \sigma_t(\mathbf{x})$ is a field, the chain rule gives:

$$\frac{\partial g_{\alpha\beta}}{\partial x^\mu} = \frac{\partial g_{\alpha\beta}}{\partial \sigma_t} \, \nabla_\mu \sigma_t. \tag{8}$$

Substituting (8) into the definition of the Christoffel symbol $\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})$

yields the exact QGD connection:

$$\boxed{\Gamma^{\lambda}_{\mu\nu}[\sigma_t] = \frac{1}{2}\, g^{\lambda\rho}\left[\frac{\partial g_{\rho\nu}}{\partial \sigma_t}\nabla_\mu \sigma_t + \frac{\partial g_{\rho\mu}}{\partial \sigma_t}\nabla_\nu \sigma_t - \frac{\partial g_{\mu\nu}}{\partial \sigma_t}\nabla_\rho \sigma_t\right].} \tag{9}$$

This is the complete Levi-Civita connection, now expressed as a linear functional of $\nabla_\mu \sigma_t$ scaled by the metric's structural sensitivity $\partial g/\partial \sigma_t$. What GR calls spacetime curvature is, in QGD, the kinematic gradient flow of the σ-field.

3.2 The Advection Tensor

For the two-body problem, the total field is $\sigma_t^{\text{tot}} = \sigma_t^{(1)} + \sigma_t^{(2)}$, so $\nabla_\mu \sigma_t^{\text{tot}} = \nabla_\mu \sigma_t^{(1)} + \nabla_\mu \sigma_t^{(2)}$. Substituting into (9), the connection of the effective one-body system splits as:

$$\widetilde{\Gamma}^{\lambda}_{\mu\nu} = \mathring{\Gamma}^{\lambda}_{\mu\nu}\left[\sigma_t^{(2)}\right] + \Omega^{\lambda}_{\mu\nu}\left[\sigma_t^{(1)}, \sigma_t^{(2)}\right], \tag{10}$$

where $\mathring{\Gamma}^{\lambda}_{\mu\nu}[\sigma_t^{(2)}]$ is the background connection generated by body 2 alone (driving the standard Keplerian orbit), and the *Advection Tensor* is:

Definition 3.1 (Advection Tensor)**.**

$$\boxed{\Omega^{\lambda}_{\mu\nu} = \frac{1}{2}\, g^{\lambda\rho}\left[\frac{\partial g_{\rho\nu}}{\partial \sigma_t}\,\nabla_\mu \sigma_t^{(1)} + \frac{\partial g_{\rho\mu}}{\partial \sigma_t}\,\nabla_\nu \sigma_t^{(1)} - \frac{\partial g_{\mu\nu}}{\partial \sigma_t}\,\nabla_\rho \sigma_t^{(1)}\right].} \tag{11}$$

Remark 3.2. The Advection Tensor is *exact*—it follows from the chain rule applied to $g_{\mu\nu}(\sigma_t^{(1)} + \sigma_t^{(2)})$ and does not require any perturbative approximation. The factor ν appearing in the post-Newtonian force law is not intrinsic to $\Omega^{\lambda}_{\mu\nu}$ itself; it enters via the Tamir–Mino energy mapping when the two-body system is reduced to an effective one-body description (Chapter 5). The notes' representation of Ω as "$\propto \nu$" is an approximation valid only after the EOB reduction.

3.3 Explicit components in the Schwarzschild sector

For the composite Schwarzschild metric $g_{tt} = -(1-\sigma_t^2)$, $g_{rr} = 1$ (isotropic gauge), the only non-vanishing metric sensitivity is:

$$\frac{\partial g_{tt}}{\partial \sigma_t} = 2\sigma_t, \qquad \frac{\partial g_{ij}}{\partial \sigma_t} = 0. \tag{12}$$

The inverse metric components are $g^{tt} = -1/(1-\sigma_t^2)$ and $g^{rr} = 1$. From Definition 3.1, the non-vanishing Advection Tensor components for a static test body 1 ($\nabla_t \sigma_t^{(1)} = 0$) are:

$$\Omega^{t}_{tr} = \frac{1}{2}\, g^{tt}\cdot 2\sigma_t^{(2)}\cdot \nabla_r \sigma_t^{(1)} = -\frac{\sigma_t^{(2)}\,\nabla_r \sigma_t^{(1)}}{1-(\sigma_t^{(2)})^2}, \tag{13}$$

$$\Omega^{r}_{tt} = -\frac{1}{2}\, g^{rr}\cdot 2\sigma_t^{(2)}\cdot \nabla_r \sigma_t^{(1)} = -\sigma_t^{(2)}\,\nabla_r \sigma_t^{(1)}. \tag{14}$$

Substituting the single-body fields $\sigma_t^{(2)} = \sqrt{r_s^{(2)}/r}$ and $\nabla_r \sigma_t^{(1)} = -\frac{1}{2}\sqrt{r_s^{(1)}}\, r^{-3/2}$:

$$\Omega^r_{tt} = \frac{1}{2}\sqrt{\frac{r_s^{(1)}\, r_s^{(2)}}{r^3}} = \frac{1}{2}\,\frac{\sqrt{4G^2 m_1 m_2/c^4}}{r^{3/2}} \propto \frac{\sqrt{m_1 m_2}}{r^2}. \tag{15}$$

After the Tamir–Mino mapping $\sqrt{m_1 m_2} \to \mu\sqrt{\nu}\, M$, this becomes a ν-weighted correction to the Newtonian force, generating the Ladder A PN terms.

3.4 The modified geodesic equation: explicit form

Substituting (10) into the geodesic condition, the equation of motion for body 1 in the field of body 2 is:

$$\frac{d^2 x^\mu}{ds^2} + \mathring{\Gamma}^\mu_{\alpha\beta}\,\frac{dx^\alpha}{ds}\frac{dx^\beta}{ds} = -\Omega^\mu_{\alpha\beta}\,\frac{dx^\alpha}{ds}\frac{dx^\beta}{ds}. \tag{16}$$

The left side is the standard geodesic equation in the field of body 2; the right side is the Advection force—the resistance of body 1's σ-field to being "dragged" through body 2's gradient.

In the static, non-relativistic limit with $dx^r/ds \ll 1$ and $dx^t/ds \approx 1$:

$$\ddot{r} \approx -\frac{GM_2}{r^2} \underbrace{-\Omega^r_{tt}}_{\text{Advection force}} = -\frac{GM_2}{r^2} + \sigma_t^{(2)}\,\nabla_r \sigma_t^{(1)}. \tag{17}$$

At Newtonian order, the second term is $\mathcal{O}(r_s^2/r^3)$ — a higher-order PN correction — confirming that the advection force is invisible at Solar-System scales and becomes significant only in the strong-field regime.

4 Stress-Energy Conservation

4.1 The stress-energy tensors

In QGD three distinct stress-energy contributions arise from the master action:

Matter stress-energy. This is the standard $T^{\mu\nu}_{\text{matter}}$, derived from varying $\mathcal{L}_{\text{eff}}(\psi, g(\sigma))$ with respect to the metric.

σ-field stress-energy. From the kinetic term in the action, contraction of the master field equation (1) with σ^μ via the Noether procedure gives:

$$T^{\alpha\beta}_{\text{field}} = \nabla^\alpha \sigma_\mu\, \nabla^\beta \sigma^\mu - \frac{1}{2}\, g^{\alpha\beta}\,(\nabla\sigma)^2 + \kappa\ell_Q^2\text{-corrections}, \tag{18}$$

which is positive-definite in the exterior region and locally conserved.

Pais–Uhlenbeck stress-energy. The massive mode G_{m_Q} carries its own stress-energy through the fourth-order kinetic sector:

$$\Theta^{\mu\nu}_{\mathrm{PU}} \sim \kappa \ell_Q^2 \left[\nabla^2 \sigma_\rho \, \nabla^\mu \nabla^\nu \sigma^\rho - \frac{1}{2} g^{\mu\nu} \, \nabla^2 \sigma_\rho \, \nabla^2 \sigma^\rho \right]. \tag{19}$$

This tensor carries the repulsive pressure associated with the massive graviton mode and is responsible for the UV regulation of the theory near $r \sim \ell_Q$.

4.2 The full conservation law

The total stress-energy conservation is:

$$\boxed{\nabla_\nu \big(T^{\mu\nu}_{\mathrm{matter}} + T^{\mu\nu}_{\mathrm{field}} + \Theta^{\mu\nu}_{\mathrm{PU}} \big) = 0.} \tag{20}$$

In the test-particle limit ($T^{\mu\nu}_{\mathrm{matter}} = \mu \, u^\mu u^\nu \, \delta^{(3)}$), equation (20) reduces to the modified geodesic equation (16), with the divergence of $T^{\mu\nu}_{\mathrm{field}}$ supplying the Advection force on the right side.

In the quantum-gravity regime ($r \sim \ell_Q$), the $\Theta^{\mu\nu}_{\mathrm{PU}}$ term contributes a pressure that prevents the total stress-energy from diverging at a point: the conservation law (20) holds everywhere, including at $r = 0$, with $\Theta^{\mu\nu}_{\mathrm{PU}}$ providing the necessary compensating pressure.

4.3 Reduction to standard GR

Proposition 4.1 (GR limit). *In the limits $\ell_Q \to 0$ and $|\sigma| \ll 1$ (weak field):*

1. $\Theta^{\mu\nu}_{\mathrm{PU}} \to 0$, *so matter and field stress-energies are conserved separately.*
2. $\Omega^\lambda_{\mu\nu} \to 0$, *recovering the standard geodesic equation.*
3. $T^{\mu\nu}_{\mathrm{field}} = T^{\mu\nu}_{\mathrm{GW}}$ *(standard gravitational wave stress-energy tensor).*
4. *Equation* (20) *reduces to* $\nabla_\nu T^{\mu\nu} = 0$ *(Einstein's conservation law).*

The corrections are suppressed by $\kappa \ell_Q^2 / r^2 \sim 10^{-70}\,\mathrm{m}^{-2}/r^2$ at astrophysical scales, consistent with all current tests of GR.

5 Comparison: GR versus QGD Equations of Motion

Feature	**General Relativity**	**QGD**
Fundamental law	Geodesic equation	σ-field equation
Connection	Levi-Civita $\Gamma[\partial g]$	Composite $\Gamma[(\partial g/\partial\sigma)\nabla\sigma]$
Advection term	Absent	$\Omega^{\lambda}_{\mu\nu}[\sigma_1, \sigma_2]$
Particle nature	Test mass on background	Source of the σ-field
Stress conservation	$\nabla_\nu T^{\mu\nu}_{\rm m} = 0$	$\nabla_\nu (T_{\rm m} + T_{\rm field} + \Theta_{\rm PU})^{\mu\nu} = 0$
High-curvature limit	Singular at $r = 0$	Regulated by $\Theta^{\mu\nu}_{\rm PU}$
Force law (radial)	$-GM/r^2 \times [1 + \mathrm{PN}]$	$-GM/r^2 \times [1 - 3\nu u^2 + \frac{63707}{720}\nu u^3]$
Weak-field limit	Newton's law	Newton's law (identical)

6 The QGD Force Law

6.1 From the connection to the radial force

The radial component of the modified geodesic equation (16), projected along the separation vector, gives the effective radial force law. In the post-Newtonian expansion, using $u \equiv GM/rc^2$ as the small parameter:

Theorem 6.1 (QGD radial force law)**.** *The effective radial acceleration in the QGD two-body system, derived from the Advection-corrected geodesic equation and expressed in terms of the EOB potential $A(u)$ (Chapter 5), is:*

$$\boxed{a_r = -\frac{GM}{r^2}\left[1 \underbrace{- 3\nu u^2}_{\textit{Ladder A}} \underbrace{+ \frac{63707}{720}\nu u^3}_{\textit{Ladder B}}\right] + \mathcal{O}(u^4).} \tag{21}$$

Proof. The EOB potential $A(u) = 1 - 2u + 2\nu u^3 - (63707/1440)\nu u^4$ is derived from the QGD integral solution in Chapter 5. The effective radial force is $a_r = -\frac{1}{2}(GM/r^2)(dA/du)$, which gives:

$$\frac{dA}{du} = -2 + 6\nu u^2 - \frac{63707}{360}\nu u^3. \tag{22}$$

Multiplying by $-\frac{1}{2}$ yields (21). □

6.2 Physical interpretation of each term

Newtonian term (1). At $u \to 0$ (large separations), the brackets reduce to unity and the force recovers Newton's inverse-square law exactly. This is mandatory: both the Levi-Civita connection and the Advection Tensor vanish in the Newtonian limit.

Ladder A term ($-3\nu u^2$, attractive). This is the leading PN correction. The coefficient -3ν arises from the cross-source $Q_t^{\text{cross}} = 2\nabla\sigma_1 \cdot \nabla\sigma_2$, which is the σ-field analogue of the graviton self-interaction. It deepens the potential well, driving perihelion precession and the post-Newtonian inspiral. In the GR limit (G_0-only propagator), this is the standard 1PN force.

Ladder B term ($+63707\nu u^3/720$, repulsive). This term arises exclusively from the massive propagator G_{m_Q} in the QGD dual-propagator $G_0 - G_{m_Q}$. Its positive sign means it acts as a repulsive "quantum pressure" that opposes the Ladder A attraction at high curvatures. At $u \sim 0.15$ (near the ISCO), the Ladder B contribution is approximately 30% of the Ladder A term for $\nu = 1/4$.

6.3 Hierarchy of scales

The three terms in (21) become comparable at different curvatures:

- Newtonian dominates for $u \ll 1$ (all astrophysical scales).
- Ladder A is $\mathcal{O}(\epsilon^2)$ where $\epsilon = v/c = \sqrt{u}$: significant for tight binaries and LIGO/LISA sources.
- Ladder B is $\mathcal{O}(\epsilon^3)$: a 1.5PN correction relative to Ladder A, detectable in principle with next-generation detectors.

The force law (21) is thus the connecting expression between the abstract σ-field dynamics (Chapter 3) and the observable gravitational wave phasing (Chapter 5).

7 Summary

Equation	Meaning
$\Box_g(1-\kappa\ell_Q^2\Box_g)\sigma_\mu = S_\mu$	Field-level EoM: all motion encoded here
$g_{\mu\nu} = g_{\mu\nu}(\sigma_t)$	Metric compositeness: no geometry without field
$\partial_\mu g_{\alpha\beta} = (\partial g_{\alpha\beta}/\partial\sigma_t)\,\nabla_\mu\sigma_t$	Chain rule: curvature = σ-gradient flow
$\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\rho}[(\partial g/\partial\sigma)\nabla\sigma]^{\mathrm{sym}}$	QGD composite connection
$\Omega^\lambda_{\mu\nu} = \Gamma^\lambda[\sigma_{\mathrm{tot}}] - \Gamma^\lambda[\sigma_2]$	Advection Tensor: field-dragging correction
$\ddot{x}^\mu + (\Gamma^\mu + \Omega^\mu)_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$	Modified worldline equation
$\nabla_\nu(T_{\mathrm{m}} + T_{\mathrm{field}} + \Theta_{\mathrm{PU}})^{\mu\nu} = 0$	Total stress-energy conservation
$a_r = -(GM/r^2)[1 - 3\nu u^2 + (63707/720)\nu u^3]$	QGD force law (to 4PN)

What is established.

- The QGD connection is derived exactly from the particle action via the composite-metric chain rule (8).
- The Advection Tensor $\Omega^\lambda_{\mu\nu}$ is defined precisely (Definition 3.1) and its components computed explicitly (equations (13)–(15)).
- The ν-dependence in the force law does not originate from Ω itself but from the Tamir–Mino EOB energy mapping.
- The three stress-energy contributions and their conservation law are identified (equation (20)).
- The GR limit is recovered exactly (Proposition 4.1).